

SYED FARHAN AHMAD, Ph.D.
Curriculum Vitae

Postdoctoral Research Associate

Xu Lab
Center of Excellence in Leukemia Studies (CELS)
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Dedicated research scientist working as Postdoctoral Research Associate at St. Jude Children's Research Hospital with expertise in repetitive DNA biology, transposable elements, satellite DNA, genome evolution, and cancer genomics. My research integrates next-generation sequencing, genome assembly, comparative genomics, transcriptomics, epigenomics, and genome-wide variation analyses to study the non-coding genome in vertebrate evolution and human disease.

I have authored or co-authored more than 65 peer-reviewed publications with over 1000 citations, spanning genetics, genomics, cell biology, aquaculture, biodiversity, conservation, and cancer biology. My research focuses on transposable element dysregulation in leukemia and other cancers, with potential relevance to tumor biology, immune signaling, therapy response, and future diagnostic and therapeutic strategies.

EDUCATION

2015-2019	Ph.D. (Biological sciences, Genetics), Sao Paulo State University, Brazil
2011-2013	MS (Biotechnology), IIUI
2005-2010	BS (Biotechnology), University of Malakand

RESEARCH TRAINING AND APPOINTMENTS

2023-present	Postdoctoral Research Associate, St. Jude Children's Research Hospital, United States
2022-2023	Lecturer, Kasetsart University, Thailand
2020-2022	Postdoctoral researcher, Kasetsart University, Thailand
2018-2019	Visiting Ph.D. fellow, University of Connecticut, United States
2015-2019	PhD Researcher (FAPESP) Sao Paulo State University, Brazil

AWARDS AND RESEARCH FUNDING

2025	ASH Abstract Achievement Award
2021	Best Postdoc presentation award, International Chromosome Conference (ICC & ICACG)
2020	Postdoc research grant Kasetsart University
2018	FAPESP research exchange fellowship
2015	FAPESP PhD fellowship
2014	Selected for TWAS postgraduate fellowships

SELECTED PUBLICATIONS (PEER REVIEWED ARTICLES)

- Jehangir M, **Ahmad SF**, Oliveira JIN, Cardoso AL, Wolf IR, Valente GT, Srikulnath K, Martins C. Genome assembly of *Astatotilapia latifasciata* uncovers B chromosome-linked chromatin reorganization. *Heredity*. 2026. PMID: 42098286.
- Lee M Jr, **Ahmad SF**, Xu J. Regulation and function of transposable elements in cancer genomes. *Cell Mol Life Sci*. 2024;81(1):157. PMID: 38556602.
- Jehangir M, **Ahmad SF**†, Singchat W, Panthum T, Thong T, Aramsirirujwet P, Lisachov A, Muangmai N, Han K, Koga A, Duengkae P, Srikulnath K. Hi-C sequencing unravels dynamic three-dimensional chromatin interactions in muntjac lineage: insights from chromosome fusions in Fea's muntjac genome. *Chromosome Res*. 2023;31(4):34. PMID: 38017297. †Equal first author.
- **Ahmad SF**, Jehangir M, Srikulnath K, Martins C. Fish genomics and its impact on vertebrate biology. *Rev Fish Biol Fisheries*. 2022;32:357–385. doi: 10.1007/s11160-021-09691-7.
- Chetruengchai W, Singchat W, Srichomthong C, Assawapitaksakul A, Srikulnath K, **Ahmad SF***, Phokaew C*, Shotelersuk V. Genome of *Varanus salvator macromaculatus* (Asian water monitor) reveals adaptations in the blood coagulation and innate immune system. *Front Ecol Evol*. 2022;10:850817. doi: 10.3389/fevo.2022.850817. *Equal corresponding author.
- Singchat W, **Ahmad SF**†, Jaisamut K, Panthum T, Ariyaphong N, Kraichak E, Muangmai N, Duengkae P, Payungporn S, Malaivijitnond S, Srikulnath K. Population-scale analysis of centromeric satellite DNA reveals highly dynamic evolutionary patterns and genomic organization in long-tailed and rhesus macaques. *Cells*. 2022;11(12):1953. PMID: 35741082. †Equal first author.
- Srikulnath K, **Ahmad SF**, Singchat W, Panthum T. Why do some vertebrates have microchromosomes? *Cells*. 2021;10(9):2182. PMID: 34571831.
- **Ahmad SF**, Singchat W, Jehangir M, Suntronpong A, Panthum T, Malaivijitnond S, Srikulnath K. Dark matter of primate genomes: satellite DNA repeats and their evolutionary dynamics. *Cells*. 2020;9(12):2714. PMID: 33352976.
- Singchat W, **Ahmad SF**, Sillapaprayoon N, Muangmai S, Baicharoen C, Indananda P, Duengkae S, Peyachoknagul RE, O'Connor R, Griffin D, Srikulnath K. Partial amniote sex chromosomal linkage homologies shared on snake W sex chromosomes support the ancestral super-sex chromosome evolution in amniotes. *Front Genet*. 2020;11:948. doi: 10.3389/fgene.2020.00948.
- **Ahmad SF**, Jehangir M, Cardoso AL, Wolf IR, Margarido VP, Cabral-de-Mello DC, O'Neill R, Valente GT, Martins C. B chromosomes of multiple species have intense evolutionary dynamics and accumulated genes related to important biological processes. *BMC Genomics*. 2020;21(1):656. doi: 10.1186/s12864-020-07072-1. PMID: 32967626.

- Ahmad SF, Singchat W, Jehangir M, Panthum T, Srikulnath K. Consequence of Paradigm Shift with Repeat Landscapes in Reptiles: Powerful Facilitators of Chromosomal Rearrangements for Diversity and Evolution. *Genes*. 2020;11(7):827. PMID: 32708239
- Jehangir M, **Ahmad SF**, Cardoso AL, Ramos E, Valente GT, Martins C. De novo genome assembly of the cichlid fish *Astatotilapia latifasciata* reveals a higher level of genomic polymorphism and genes related to B chromosomes. *Chromosoma*. 2019;128(2):81–96. PMID: 31115663.
- **Ahmad SF**, Martins C. The modern view of B chromosomes under the impact of high-scale omics analyses. *Cells*. 2019;8(2):156. PMID: 30781835.
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ABSTRACTS AND CONFERENCES PRESENTATIONS

- **Ahmad SF**, Wang K, Hu W, Lee MJ, Zhang Y, Ma J, Ribeiro RC, Pounds S, Klco JM, Xu J. Elucidating the role of retrotransposon-mediated viral mimicry in leukemia therapy. *67th ASH Annual Meeting and Exposition*. December 8, 2025. Oral presentation. *Blood*. 2025;146(Suppl 1):774. doi: 10.1182/blood-2025-774.
- **Ahmad SF**, Wang K, Hu W, Lee MJ, Zhang Y, Ma J, Ribeiro RC, Pounds S, Klco JM, Xu J. Decoding regulatory roles and rules of retrotransposon-mediated viral mimicry during hypomethylation therapy. *Cancer Center Postdoctoral Symposium*. February 20, 2026. Abstract presentation.
- **Ahmad SF**, Wang K, Hu W, Lee MJ, Zhang Y, Ma J, Ribeiro RC, Pounds S, Klco JM, Xu J. Understanding the roles and rules of immunogenic retrotransposon-mediated viral mimicry in leukemia therapy. *Pathology Research Symposium*. April 25, 2026. Oral presentation.
- **Ahmad SF**, Wang K, Hu W, Lee MJ, Zhang Y, Ma J, Klco JM, Xu J. Elucidating the role of genomic repetitive element-mediated viral mimicry response in leukemia therapy. *Pathology Research Symposium*. April 25, 2025. Poster presentation.
- **Ahmad SF**, Srikulnath K. Siamese cobra genome and evolutionary trajectories of chromosome rearrangements. *23rd International Chromosome Conference and 24th International Colloquium in Animal Cytogenetics and Genomics*. July 13–17, 2021. Oral presentation.
- **Ahmad SF**. SNP detection, HRM analysis, and genotyping using real-time PCR. *Online Workshop: Basic Real-Time PCR Technique for Gene Quantification and Detection*. November 8–22, 2021. Invited speaker.
- **Ahmad SF**, Malaivijitnond S, Srikulnath K. Population genomics analysis provides new insights into genetic admixture and diversity of rhesus and long-tailed macaques. *1st Symposium of the Natural History Museum, National Science Museum Thailand*. September 24–25, 2020. Invited speaker.
- **Ahmad SF**, Martins C. The impact of omics analyses in the modern view of B chromosomes. *4th B Chromosome Conference*. Botucatu, Brazil. July 20–23, 2019. Oral presentation. *BMC Proc*. 2019;13. doi: 10.1186/s12919-019-0165-x.
- **Ahmad SF**, Cabral-de-Mello DC, Parise-Maltempi P, Margarido VP, O'Neill R, Valente GT, Martins C. The integrative genomics and bioinformatics analysis unveil the genetic contents and evolutionary origin of B chromosomes. *Conference on Regulatory and Systems Genomics, International Society for Computational Biology*. New York, NY, USA. 2018. Poster presentation.
- Martins C, Nakajima RT, Fantinatti B, Marques DF, Venturelli NB, Jehangir M, Ramos E, Oliveira JI, **Ahmad SF**, Cardoso A, Valente GT. B chromosomes in the light of functional analysis. *23rd International Colloquium of Animal Cytogenetics and Genomics*. Saint Petersburg, Russia. 2018. *Comp Cytogenet*. 2018;12:299–360. doi: 10.3897/CompCytogen.v12i3.274.

- **Ahmad SF**, Cabral-de-Mello DC, Parise-Maltempi P, Margarido VP, O'Neill R, Valente GT, Martins C. Fishing for B chromosome genes. *XVIII Simpósio de Citogenética e Genética de Peixes*. Cascavel, Brazil. 2018. Poster presentation.
- Martins C, Nakajima RT, Ramos E, Fantinatti B, Valente GT, Cardoso A, Jehangir M, Marques DF, Oliveira JI, Bovolenta IA, Venturelli NB, Kocher TD, **Ahmad SF**. Fishing for B chromosome effects. *22nd International Chromosome Conference*. Prague, Czech Republic. 2018.
- **Ahmad SF**, Valente GT, Martins C. The distribution of repetitive DNAs between regular and super-numerary chromosomes in the cavefish genome. *36th International Society for Animal Genetics Conference*. University College Dublin, Ireland. 2016. Poster presentation.
- **Ahmad SF**, Jehangir M, Cardoso A, Martins C. FISH mapping of 45S rRNA and *Ihbb* genes on autosomes and B chromosome of cichlid fish *Astatotilapia latifasciata*. *5th Brazilian Meeting of Cytogenetics and Cytogenomics*. Londrina, Brazil. 2017. Poster presentation.
- Jehangir M, **Ahmad SF**, Valente GT, Martins C. A coverage-based study of *Ihbb* and 45S rRNA genes in the B chromosomes of *Astatotilapia latifasciata* to highlight polymorphism levels. *21st International Chromosome Conference*. Foz do Iguaçu, Brazil. 2016. *Cytogenet Genome Res.* 2016;148:83–155. doi: 10.1159/000446523.
- **Ahmad SF**, Valente GT, Jehangir M, Coan R, Martins C. Genomic analysis of transposable elements in *Astyanax correntinus* with focus on B chromosome. *21st International Chromosome Conference*. Foz do Iguaçu, Brazil. 2016. Poster presentation. *Cytogenet Genome Res.* 2016;148:83–155. doi: 10.1159/000446523.
- Jehangir M, **Ahmad SF**, Valente GT, Ramos E, Fantinatti B, Martins C. Assembly of the cichlid fish *Astatotilapia latifasciata* genome with focus on B chromosome. *61st Congresso Brasileiro de Genética*. Águas de Lindoia, Brazil. 2015.
- **Ahmad SF**, Valente GT, Martins C. Searching for transposable elements in the genome of cavefish *Astyanax mexicanus*. *61st Congresso Brasileiro de Genética*. Águas de Lindoia, Brazil. 2015. Poster presentation.

TEACHING

Bioscience 01416421 Human genetics
 Bioscience 01444491 Research Methods in Bioscience and Technology
 Bioscience 01416454 Introduction to Bioinformatics

SERVICE

Editor: BMC Genomics, Comparative Cytogenetics
Reviewer: Genes and Genomics, BMC Genomics, Plos Computational Biology, Frontiers in Genetics, Life, Biomedicines, Human Gene, Genes, Communications Biology
Guest editor: Special Issue (Sex chromosome biology and genome evolution)
Topic Coordinator: Frontiers in Genetics
Postdoc member: Genetics Society of America
Organizer: The International Conference of the Genetics Society of Thailand 2022 (iGST2022)
Instructor: Workshop Basic Bioinformatics for Biologist

Moderator: Special seminars in Genome informatics and Bioresource for Animals
Thesis Committee: PhD program in Biosciences, Faculty of science, Kasetsart University
Advisor: Two undergraduate students in Biosciences and Technology Program
Co-advisor: Two PhD students doing research at Animal Genomics and Bioresource Center

RESEARCH SKILLS AND EXPERTISE

Bioinformatics and Computational Genomics:

- Expertise in use of HPC and servers
- Strong background in programming and algorithms Bash (linux), Python and R for processing, and statistical analysis of NGS short and long reads data
- Deep theoretical and practical knowledge of long reads (PacBio, Nanopore) and short reads (Illumina) sequencing and data management, quality evaluation and data analysis skills with bioinformatics
- Comparative genomic (Whole genome alignments, synteny analysis)
- Genome assemblies and annotations
- Identification of genetic polymorphism and mutations, Genome-wide SNPs and genotyping
- Structural variations and rearrangements analysis of genomes
- Population genomics, phylogenomics and statistical genetics
- Experienced in bioinformatic tools, including Bowtie 2, BWA, minimap2, Circos, Samtools, GATK, Bedtools, MUGSY, MUSCLE, SoapDenovo, Velvet, MAKER, BLAST2GO, VCFtools, Canu, FASTXtoolkit, Stacks, Populations, Demographic analysis, Orthofinder, CAFÉ, Gene Expressions Analysis tools, visualization tools, Gene Analysis tools, prediction tools, GWAS packages like PLINK etc
- TEtranscripts, Telescope, SQuIRE, SalmonTE, RepEnrich, RepeatMasker, RepeatModeler, RepeatExplorer, Dfam, Repbase-based annotation, featureCounts/HTSeq-based TE quantification, and custom TE-aware pipelines
- Clinical Genomics, large scale patients' genomes and transcriptomes analysis
- Epigenomics ATAC-seq, WGBS and ChIP-seq data analysis
- Development of pipelines for transposable elements expression profiles
- Development of Patient response score to epigenetic therapy
- Infographics design: Photoshop, Illustrator, InkScape and SVG production

Molecular Biology and Cytogenetics:

- DNA/RNA extraction
- Gel electrophoresis
- Sequencing libraries preparation
- qPCR
- Florescent Insitu Hybridization (FISH)